C 语言程序设计重点习题代码（自动化 2201）

P18　例 2.3 判断闰年：

```
#include<stdio.h>
int main()
{
	for (int i = 2000; i < 2501; i++)
	{
		if ((i % 4 == 0 && i % 100 != 0) || (i % 400 == 0))
		{
			printf("%8d", i);
		}
	}
	return 0;
}
```

P79　例 3.10 大写转小写：

```
#include<stdio.h>
int main()
{
	char c1, c2;
	c1 = getchar();
	c2 = c1 + 32;
	putchar(c2);
	return 0;
}
```

P122 例 5.4 break 语句.宏定义：

```
#include<stdio.h>
#include<string.h>
#define N 10
char str[N];
int main()
{
	void sort(char[]);
	int i, flag;
	printf("input string:\n");
	for (flag =0; flag < N; flag++)
	{
		scanf_s("%c", &str[flag]);
	}
	sort(str);
	printf("string sorted:\n");
	for (i = 0; i < N; i++)
	{
		printf("%c", str[i]);
	}
	printf("\n");
	return 0;
}

void sort(char str[])
{
	int i, j;
	char t;
	for (j = 1; j < N; j++)
	{
		for (i = 0; (i < N - j) && (str[i] != '\0'); i++)
		{
			if (str[i] > str[i + 1])
			{
				t = str[i];
				str[i] = str[i + 1];
				str[i + 1] = t;
			}
		}
	}
}
```

P127 例 5.7 求 Π 算法：

```
#include<stdio.h>
#include<math.h>
int main()
{
	int sign = 1;
	float n = 1, a = 1, sum = 0;
	while (fabs(a) >= 1e-6)
	{
		sum = sum + a;
		n = n + 2;
		sign = sign *( - 1);
		a = sign / n;
	}
	sum = 4 * sum;
	printf("pi=%10.8f", sum);
```

```
    return 0;
}
```
P129 例 5.8 斐波那契数列求兔子数:
```
#include<stdio.h>

int main()
{
    int f1 = 1, f2 = 1, f3;
    printf("%d\n%d\n", f1, f2);
    for (int i = 0; i < 38; i++)
    {
        f3 = f1 + f2;
        printf("%d\n", f3);
        f1 = f2;
        f2 = f3;
    }
    return 0;
}
```
改进:
```
#include<stdio.h>
int main()
{
    int f1 = 1, f2 = 1;
    for (int i = 1; i < 21; i++)
    {
        printf("%12d%12d", f1, f2);
        if (i % 2 == 0)
        {
            printf("\n");
        }
        f1 = f1 + f2;
        f2 = f1 + f2;
    }
    return 0;
}
```
P134 例 5.10 判断素数并输出:
```
#include<stdio.h>
int main()
{
    int m = 0;
    for (int i = 101; i < 200; i = i + 2)
    {
        bool f = true;
        for (int j = 2; j < i; j++)
        {
            if (i % j == 0)
            {
                f = false;
                break;
            }
        }
        if (f == true)
        {
            printf("%4d", i);
            m++;
            if (m % 5 == 0)
            {
                printf("\n");
            }
        }
    }
    return 0;
}
```
P144 例 6.3 起泡法排序:
```
#include<stdio.h>
int main()
{
    int a[10];
    int i, j, t;
    for (i = 0; i < 10; i++)
    {
        scanf_s("%d", &a[i]);
    }
    printf("\n");
    for (i = 0; i < 9; i++)
    {
        for (j = 0; j < 9-i; j++)
        {
            if (a[j] > a[j + 1])
            {
                t = a[j];
                a[j] = a[j + 1];
                a[j + 1] = t;
            }
        }
    }
    for (i = 0; i < 10; i++)
    {
```

```
            printf("%5d",a[i]);
        }
        return 0;
}
```

P150 例 6.5 二维数组找最大值（打擂台算法）：

```
#include<stdio.h>
int main()
{
    int a[3][4];
    int i, j, c = 0, r = 0, max;
    for (i = 0; i <3; i++)
    {
        for (j = 0; j < 4; j++)
        {
            scanf_s("%d", &a[i][j]);
        }
    }
    max = a[0][0];
    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 4; j++)
        {
            if (a[i][j] > max)
            {
                max = a[i][j];
                r = i, c = j;
            }
        }
    }
    printf("max=%d\n 第 %d 行 第 %d 列 \n",max,r,c);
    return 0;
}
```

P163 例 6.8 统计单词数：

```
#include <stdio.h>
int main()
{
    char a[100];
    int i, word, num;
    gets_s(a);
    num = 0;
    word = 0;
    for (i = 0; a[i]!='\0'; i++)
    {
        if (a[i] == ' ')// 检测到空格
        {
            word = 0; // 设置标记位为不在单词内
        }
        else if (word == 0)// 在单词内
        {
            num++; // 统计单词个数
            word = 1; // 设置标记位为在单词内
        }
    }
    printf("单词数为%d", num);
    return 0;
}
```

P173 例 7.2 调用函数输出最大值：

```
#include <stdio.h>
int main()
{
    int max(int a, int b);
    int a, b;
    scanf_s("%d %d", &a, &b);
    printf("max=%d\n", max(a, b));
}
int max(int a, int b)
{
    int max = a;
    if (b > max)
    {
        max = b;
    }
    return(max);
}
```

P196 例 7.10 调用数组函数输出平均值：

```
#include <stdio.h>
int main()
{
    float average(float a[]);
    float a[10];
    for (int i = 0; i < 10; i++)
    {
        scanf_s("%f", &a[i]);
    }
```

```
    printf("aver=%5.2f\n", average(a));
    return 0;
}
float average(float a[])
{
    float aver, sum = 0;
    for (int i = 0; i < 10; i++)
    {
        sum = sum + a[i];
    }
    aver = sum / 10;
    return(aver);
}
```

P193 例 7.12 函数调用选择法排序:

```
include <stdio.h>
int main()
{
    void sort(int array[], int n);
    int a[10], i;
    printf("enter array:\n");
    for (i = 0; i < 10; i++)
    {
        scanf_s("%d", &a[i]);
    }
    sort(a, 10);
    printf("The sorted array:\n");
    for (i = 0; i < 10; i++)
    {
        printf("%d\t", a[i]);
    }
    printf("\n");
    return(0);
}
void sort(int array[], int n)
{
    int i, j, temp;
    for (i = 0; i < 10; i++)
    {
        for (j = i + 1; j < 10; j++)
        {
            if (array[i] > array[j])
            {
                temp = array[j];
                array[j] = array[i];
                array[i] = temp;
            }
        }
    }
}
```

P195 例 7.13 函数调用求数组元素最大值:

```
#include <stdio.h>
int main()
{
    int max(int array[][4]);
    int a[3][4], i, j;
    printf("enter array:\n");
    for (i = 0; i < 3; i++)
    {
        for (j = 0; j < 10; j++)
        {
            scanf_s("%d", &a[i][j]);
        }
    }
    printf("The max value:%d\n",max(a));
    return 0;
}
int max(int array[][4])
{
    int i, j, max=array[0][0];
    for (i = 0; i < 3; i++)
    {
        for (j=0; j < 4 ; j++)
        {
            if (array[i][j] >max)
            {
                max=array[i][j];
            }
        }
    }
    return(max);
}
```

P198 例 7.14 使用全局变量返回多个函数值:

```
#include <stdio.h>
float Max = 0, Min = 0;
int main()
{
    float average(float a[]);
```

```
    float a[10];
    for (int i = 0; i < 10; i++)
    {
        scanf_s("%f", &a[i]);
    }
    printf("aver=%5.2f\n", average(a));
    printf("Max=%5.2f    Min=%5.2f\n", Max,
Min);
    return 0;
}
float average(float a[])
{
    float aver, sum = 0;
    Max = Min = a[0];
    for (int i = 0; i < 10; i++)
    {
        if (a[i] > Max)
        {
            Max = a[i];
        }
        if (a[i] < Min)
        {
            Min = a[i];
        }
        sum = sum + a[i];
    }
    aver = sum / 10;
    return(aver);
}
```

P224 例 8.3 利用指针做函数参数对数字按大小输出：

```
#include <stdio.h>
int main()
{
    void fun(int * p1, int * p2);
    int a, b;
    int * p1 = &a, * p2 = &b;
    printf("enter a and b:\n");
    scanf_s("%d %d", p1, p2);
    if (a < b)
    {
        fun(p1, p2);
    }
    printf("max is %d    min is %d\n", a, b);
    printf("max is %d    min is %d", *p1, *p2);
    return 0;
}
void fun(int* p1, int* p2)
{
    int t;
    t = *p1;
    *p1 = *p2;
    *p2 = t;
}
```

P238 例 8.8 数组名做函数参数对数组中的 N 个整数按相反顺序输出：

```
#include <stdio.h>
int main()
{
    void fun(int a[], int n);
    int a[10] = { 0,1,2,3,4,5,6,7,8,9 };
    printf("The original array:  \n");
    for (int i = 0; i < 10; i++)
    {
        printf("%d\t", *(a+i));
    }
    fun(a, 10);
    printf("\nThe new array:  \n");
    for (int i = 0; i < 10; i++)
    {
        printf("%d\t", *(a + i));
    }
    return 0;
}
void fun(int a[], int n)
{
    int t, i, m = (n - 1) / 2;
    for (i = 0; i <= m; i++)
    {
        t = a[i];
        a[i] = a[n - i - 1];
        a[n - i - 1] = t;
    }
}
```

P242 例 8.10 指针 函数调用 选择法排序：

```
#include <stdio.h>
int main()
{
```

```
    void sort(int array[], int n);
    int a[10], i;
    int* p = a;
    printf("enter array:\n");
    for (i = 0; i < 10; i++,p++)
    {
        scanf_s("%d", p);
    }
    p = a;//指针变量归位
    sort(p, 10);
    printf("The sorted array:\n");
    for (i = 0; i < 10; i++)
    {
        printf("%d\t", *(a+i));
    }
    printf("\n");
    return(0);
}
void sort(int array[], int n)
{
    int i, j, temp;
    for (i = 0; i < n-1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (array[i] > array[j])
            {
                temp = array[j];
                array[j] = array[i];
                array[i] = temp;
            }
        }
    }
```

P301 例 9.4 结构体数组：重点

```
#include <stdio.h>
struct Student
{
    int num;
    char name[20];
    float score;
};
int main()
{
    struct Student student[5] =
{ {10101,"zhang san",78},{10102,"li si",99},
        {10103,"wang  liang",88},{10104,"li
ming",101},{10105,"yuan qing",317} };
    struct Student stu;
    int i, j;
    for (i = 0; i < 4; i++)
    {
        for (j = i + 1; j < 5; j++)
        {
            if(student[j].score>
student[i].score)
            {
                stu = student[j];
                student[j] = student[i];
                student[i] = stu;
            }
        }
    }
    for (i = 0; i < 5; i++)
    {
        printf("%6d%12s%10.2f\n",
student[i].num,          student[i].name,
student[i].score);
    }
    printf("\n");
    return 0;
}
```

课后习题：

P137.8 水仙花数：

```
#include <stdio.h>
int main()
{
    int i;
    for (i = 100; i < 1000; i++)
    {
        int x = i / 100;
        int y = (i - 100 * x) / 10;
        int z = i - 100 * x - 10 * y;
        if (i == x * x * x + y * y * y + z * z *
z)
        {
            printf("%5d", i);
```

```
            }
        }
        return 0;
}
```

P165.2 选择法排序:

P166.13 My_strcat 函数:

```
#include <iostream>
void main( )
{
        char str0[41], str1[21];
        int j=0, k=0;
        printf("Please input first 1-20 characters:\n");
        scanf_s("%s", str0,41);
        printf("Please input second 1 - 20 characters:\n");
        scanf_s("%s", str1,20);
        for (j = 0; str0[j] != '\0'; j++);
        for (k = 0; str1[j] != '\0'; k++,j++)
        {
                str0[j] = str1[k];
        }
        str0[40] = '\0';
        printf("%s\n", str0);
}
```

P215.3 判断素数:

```
#include <iostream>
int main()
{
        int a;
        bool f = true;
        printf("please enter a number :\n");
        scanf_s("%d", &a);
        for (int i = 2; i < a; i++)
        {
                if (a % i == 0)
                {
                        f = false;
                        break;
                }
        }
        if (f == true)
        {
                printf("%d 是一个素数", a);
        }
        else
        {
                printf("%d 不是一个素数", a);
        }
        return 0;
}
```

P216.7

原音字母筛选:

```
#include <iostream>
int main()
{
        char a[100], b[100]={};
        int j = 0;
        scanf_s("%s", a,100);
        for (int i = 0; i < 100; i++)
        {
                if (a[i] == 'a' || a[i] == 'e' || a[i] == 'i' || a[i] == 'o' || a[i] == 'u'
                        || a[i] == 'A' || a[i] == 'E' || a[i] == 'I' || a[i] == 'O' || a[i] == 'U')
                {
                        b[j] = a[i];
                        j++;
                }
        }
        b[j] = '\0';
        printf("%s", b);
        return 0;
}
```

P216 习题 11 调用函数用起泡法对 10 个字符排序:

```
#include<stdio.h>
#include<string.h>
#define N 11
int main()
{
        void sort(char[]);
        int i, flag;
        char str[N];
        printf("input string:\n");
        scanf_s("%s",str,N);
        sort(str);
```

```
	printf("string sorted:\n");
	for (i = 0; i < N; i++)
	{
		printf("%c", str[i]);
	}
	printf("\n");
	return 0;
}

void sort(char str[])
{
	int i, j;
	char t;
	for (j = 1; j < N; j++)
	{
		for (i = 0; (i < N - j) && (str[i] != '\0'); i++)
		{
			if (str[i] > str[i + 1])
			{
				t = str[i];
				str[i] = str[i + 1];
				str[i + 1] = t;
			}
		}
	}
}
```

P291 第 3 题　利用指针调用函数:

```
#include <iostream>
int main()
{
	void input(int*);
	void max_min_value(int*);
	void output(int*);
	int number[10];
	input(number);
	max_min_value(number);
	output(number);
	return 0;
}
void input(int* number)
{
	int i;
	printf("input 10 numbers;");
	for (i = 0;i < 10;i++)
	{
		scanf_s("%d", &number[i]);

	}
}
void max_min_value(int* number)
{
	int* max, * min, * p, temp;
	max = min = number;
	for (p = number + 1;p < number + 10;p++)
	{
		if (*p > *max)
		{
			max = p;
		}
		else
		{
			if (*p < *min)
			{
				min = p;
			}
		}
	}
	temp = number[0];
	number[0] = *min;
	*min = temp;
	if (max == number)
	{
		max = min;
	}
	temp = number[9];
	number[9] = *max;
	*max = temp;
}
void output(int* number)
{
	int* p;
	printf("Now, they are : ");
	for (p = number;p < number + 10;p++)
	{
		printf("%d", *p);
```

```
            printf("\n");
        }
}
```

P 330 第 5 题 结构体应用：

```
#include <stdio.h>
struct Student
{
    int num;
    char name[20];
    double score1;
    double score2;
    double score3;
    double average;
};
int main()
{
    struct Student stu[10];

    int m = 0;
    for (m = 0; m < 10; m++)
    {
        printf("请输入第%d 位学生\n 学号：",m+1);
        scanf_s("%d", &stu[m].num);
        printf("姓名： ");
        scanf_s("%s",stu[m].name,20);
        printf("三门课程成绩： ");
        scanf_s("%lf %lf %lf",&stu[m].score1, &stu[m].score2, &stu[m].score3);
    }
    double sum1 = 0, sum2 = 0, sum3 = 0, average1, average2, average3;
    int n = 0;
    for (n = 0; n < 10; n++)
    {
        sum1 += stu[n].score1;
        sum2 += stu[n].score2;
        sum3 += stu[n].score3;
    }
    average1 = sum1 / 10.0;
    average2 = sum2 / 10.0;
    average3 = sum3 / 10.0;
    printf("三门课程平均成绩为： ");
    printf("%6.2lf%6.2lf%6.2lf.\n",  average1, average2, average3);
    double sum4;
    int i = 0, j = 0;
    for (i = 0; i < 10; i++)
    {
        sum4 = stu[i].score1 + stu[i].score2 + stu[i].score3;
        stu[i].average = sum4 / 3.0;
    }
    struct Student student=stu[0];
    for (j = 1; j < 10; j++)
    {
        if (stu[j].average > student.average)
        {
            student = stu[j];
        }
    }
    printf(" 最 高 分 学 生 学 号 ： %6d\n", student.num);
    printf(" 最 高 分 学 生 姓 名 ： %8s\n", student.name );
    printf(" 最 高 分 学 生 三 门 课 程 成 绩 :%6.2lf%6.2lf%6.2lf\n",student.score1, student.score2, student.score3);
    printf("6.2lf\n", student.average);
    return 0;
}
```